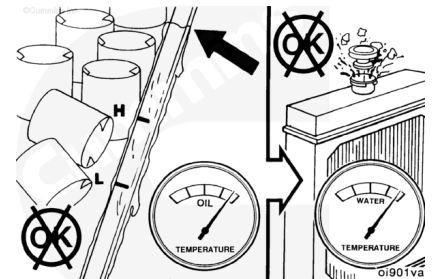


Trainings ▾

General Information

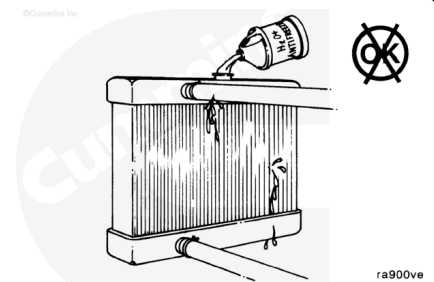
When you troubleshoot overheating, remember that too much oil in the oil pan can cause additional heat from friction when the rod journals are submerged in oil.



Overfilling with oil raises the oil temperature that is transferred to the coolant system at the oil cooler.

The system is designed to use a specific quantity of coolant. If the coolant level is low, the engine will run hot.

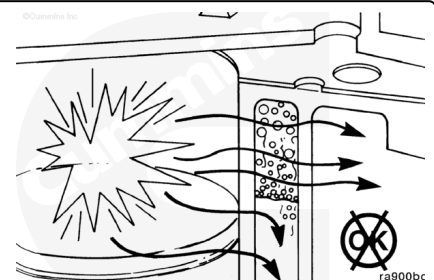
Note : The engine or system has a leak if frequent addition of coolant is necessary. Find and repair the leak.



⚠ CAUTION ⚠

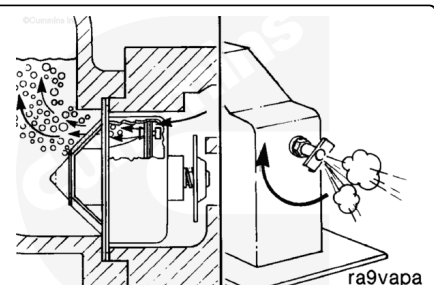
The engine coolant passages must be completely filled with coolant.

During operation, entrapped air mixes with the coolant, which results in cavitation corrosion and poor heat transfer. Highly aerated coolant can cause localized overheating of the cylinder head and block, which can result in a cracked head, scored cylinder, or blown head gasket.



During filling, air **must** be vented from the engine coolant passages. The air vents through the jiggle pin openings to the top radiator hose and out the fill opening. Additional venting is provided for engines equipped with an aftercooler. Open the petcock during filling.

Note : Adequate venting is provided for a fill rate of 14 liters per minute [3.7 gallons per minute].



Note : Excess air in the cooling system can lead to overheating and low coolant levels.

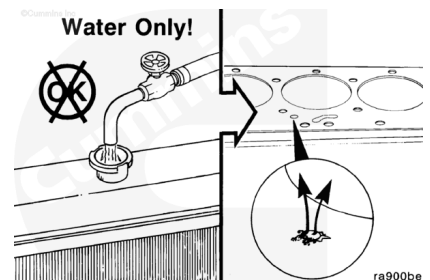
Note : A 50-percent mixture of antifreeze and water **must** be premixed before filling the system. The ability of antifreeze to remove heat from the engine is **not** as good as water, so pouring antifreeze into the engine first could contribute to an overheated condition before the liquids are completely mixed.

A mixture of 50-percent ethylene-glycol base antifreeze is required for operation of the engine in temperature environments above -37°C [-35°F]. A mixture of 40-percent water and 60-percent antifreeze is recommended for temperatures below -37°C [-35°F]. **Never** use more than 60-percent antifreeze.



⚠ CAUTION ⚠

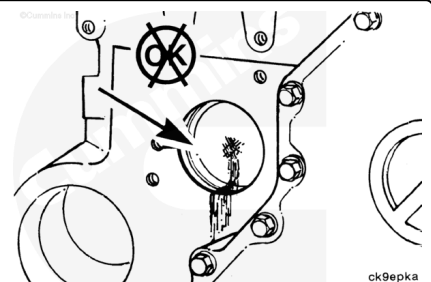
Never use water alone for coolant. Damage from corrosion can be the result of using water alone for coolant. The small holes in the head gasket are especially susceptible to plugging. These holes are orifices and their size is critical. Do not enlarge the size of the orifices. To do so will disturb the coolant flow and will not solve an overheating problem.



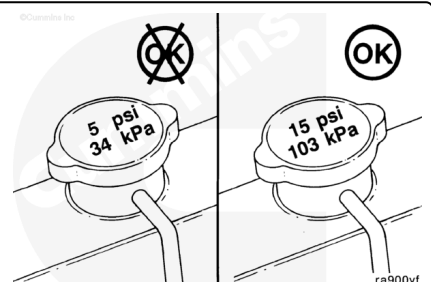
Water will cause rust formation, reducing the flow in the smaller coolant passages.

Also, water used as a coolant for even a relatively short period can result in the cup plugs rusting through, allowing the coolant to leak.

Note : A sudden loss of coolant from a heavily loaded engine can result in severe damage to the pistons and cylinder bore.

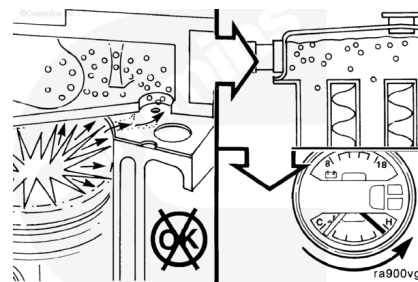


An incorrect or malfunctioning cap can result in the loss of coolant and the engine running hot. Refer to Procedure 008-047 ([/qs3/pubsys2/xml/en/procedures/40/40-008-047.html](http://qs3/pubsys2/xml/en/procedures/40/40-008-047.html)).



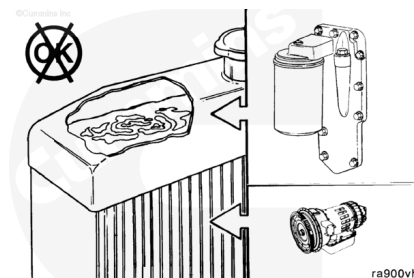
Air in the coolant can result in loss of coolant from the overflow when the aerated coolant is hot. The heated air expands, increasing the pressure in the system, causing the cap to open.

Similarly, coolant can be displaced through the overflow if the head gasket leaks compression gases to the coolant system.



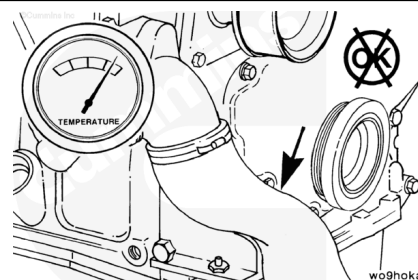
The operating pressure of the coolant system and the lubricating system can result in the mixing of the fluids if there is a leak between the systems: head gasket, oil cooler, etc. (refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>), Lubricating Oil System in Section 7).

Note : Transmission fluid can also leak into the coolant through radiator bottom tank transmission oil coolers. Refer to the OEM service manual.

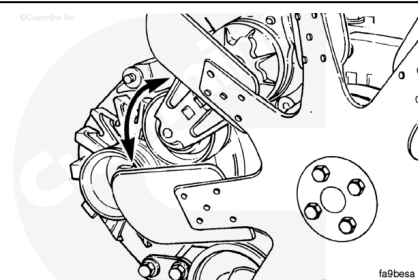


Water Pump

The water pump pulls coolant from the bottom of the radiator and pumps it through the engine back to the top of the radiator for cooling. Reduced or interrupted flow will result in the engine running hot.

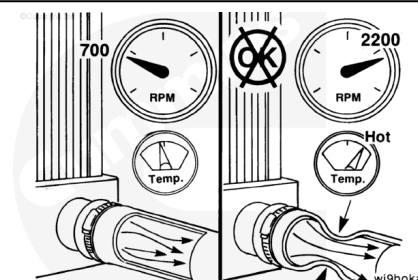


The pump is belt-driven from the crankshaft pulley. An automatic belt tensioner is used to prevent the belt from slipping on the pump pulley. A malfunction of the tensioner will cause the water pump impeller to rotate at a slower speed, reducing the amount of coolant flow.

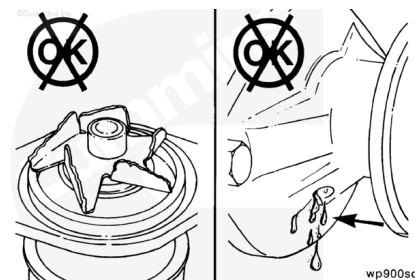


The coolant flow can also be reduced if the inlet hose to the water pump collapses. A hose will usually **not** collapse while the engine is running at low speed. Check the hose while the engine is running at rated speed.

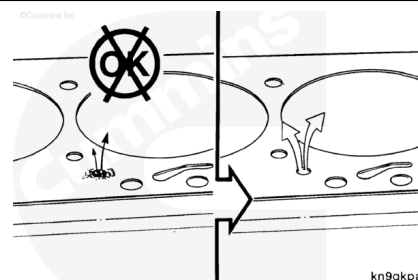
Note : Be sure the engine is warm, a minimum of 95°C [203°F], so the thermostat is open.



A worn or malfunctioning water pump will **not** produce the flow required to prevent the engine from running hot. However, be sure to check the other possibilities indicated in the Troubleshooting Logic before checking the flow or replacing the pump. Refer to Procedure 008-062 ([/qs3/pubsys2/xml/en/procedures/40/40-008-062-tr.html](http://qs3/pubsys2/xml/en/procedures/40/40-008-062-tr.html)).

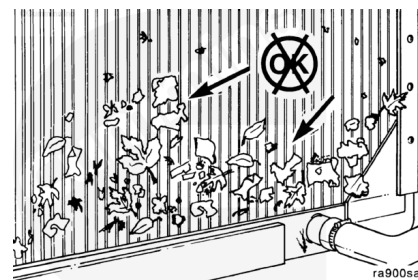


As stated in the coolant discussion, an obstruction in the passages can interrupt flow.



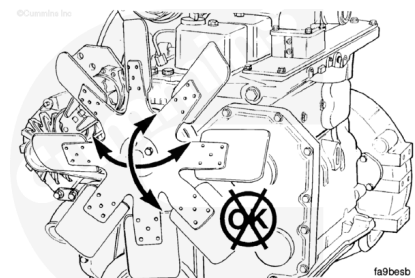
Radiator, Fans, and Shutters

Air forced through the fins of the radiator by a fan cools the coolant pumped through the radiator. Environmental debris (paper, straw, lint, dust, etc.) can obstruct the fins and stop the flow of air, which will reduce the cooling effect of the radiator.



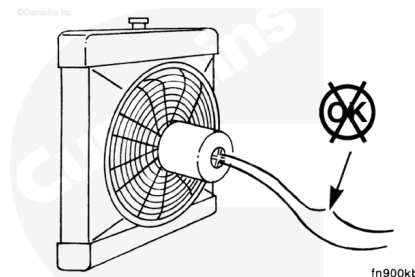
If the fan is belt-driven, a slipping belt will result in slower fan speed and reduced cooling. A malfunctioning automatic belt tensioner can be the problem.

Note : Check the bearings in the fan hub and other pulleys to make sure they are **not** causing excessive belt vibration and slippage.



Interruption of the circuit to an electrically driven fan can result in insufficient airflow and cause the engine to run hot.

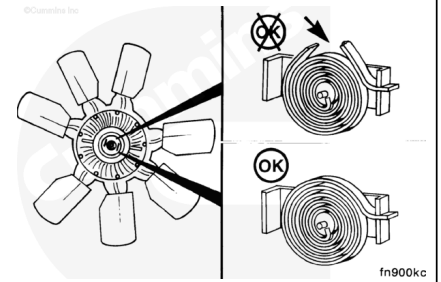
Note : Make sure the temperature sensor is functioning correctly.



Some applications use thermatic fans. These fans operate **only** as needed to keep the coolant at the correct temperature. If the fan does **not** operate when the coolant temperature

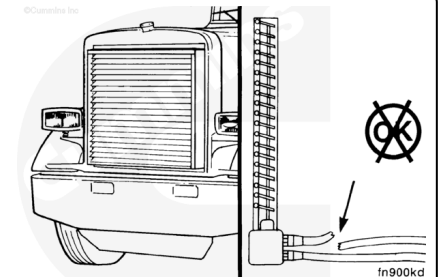
increases, the engine will run hot. If the fan does **not** shut off when coolant temperature decreases, the engine will run cold.

Note : Make sure the coolant temperature sensor is functioning correctly.

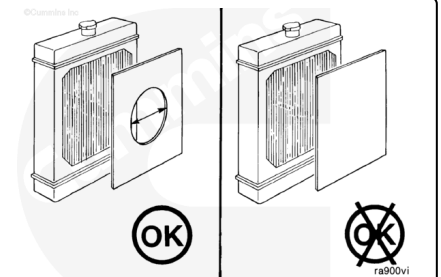


Shutters are designed to control airflow across the radiator. If the shutters fail to open when needed, the engine can run hot. Failure of the shutters to close can result in too much airflow and the engine running cold.

Note : Make sure that the air temperature sensor is functioning correctly. Check the air-operated shutter controls. Check for air leaks.

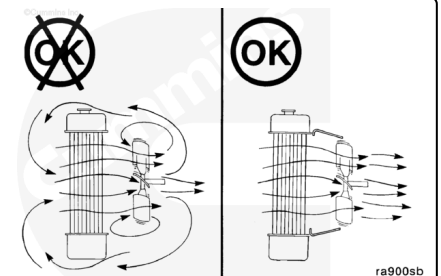


Winterfronts can be used on a charge air cooled engine, but **must** be designed to cover the frontal area of the cooling system partially. A minimum of 120 square inches of charge air cooled frontal area **must** be left open to airflow.



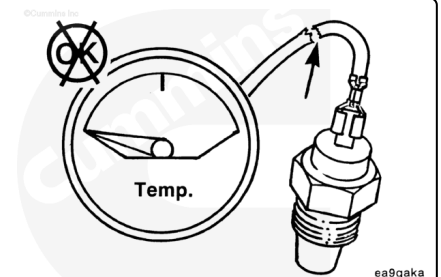
An incorrect fan shroud or obstructions can reduce airflow and cause the engine to run hot.

Note : Check to be sure air is not recirculating. Check for missing baffles.



Gauges, Overfueling, and Loading

Gauges and sensors are used in the system to measure the coolant temperature. These can malfunction and provide an incorrect temperature indication.



CAUTION

Overfueling can cause the engine to overheat. Make sure that the fuel pump is calibrated correctly.



⚠ CAUTION ⚠

Constant overloading (lugging) can cause the engine to run hot.

